I、 Data Acquisition

1. Data Acquisition System

This study uses the Texas Instruments TI-IWR1843 77 GHz FMCW millimeter-wave radar for wrist pulse wave extraction, configured with 3 transmitters and 4 receivers to form 12 virtual channels. A blood pressure monitor synchronously measures reference systolic blood pressure (SBP) and diastolic blood pressure (DBP) for model training.

Operating at 77 GHz with a 3.9 mm wavelength, the radar provides high phase sensitivity for micrometer-level wrist micro-displacement detection, enabling accurate pulse wave signal recovery from reflected signal phase variations..

2. Data Acquisition Environment and Procedure

(1) Data Acquisition Environment

The data acquisition environment in this study is illustrated in the Fig.1.

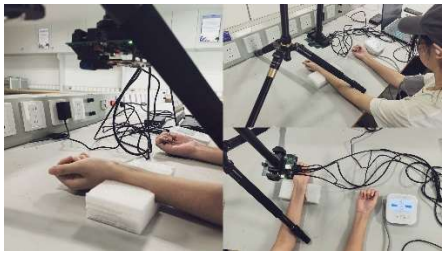

Fig 1 Data acquisition environment

The radar was placed 15 cm from the wrist for high SNR and stability. A sponge pad supported and fixed the wrist to suppress body movement interference and expose the wrist artery, enhancing pulse wave micro-motion detection. A standard electronic sphygmomanometer measured systolic (SBP) and diastolic blood pressure (DBP) synchronously, establishing one-to-one radar-BP signal pairs.n.

(2) Data Acquisition Procedure

a) Record subject ID, height, weight, and hypertension history.
b) Collect radar pulse wave signals and synchronous blood pressure..
c) Save radar data, BP data, measurement time, wrist side, and posture.

(3) Data Acquisition Results

Data were collected from 16 subjects (over 300 valid samples). After screening abnormal/insufficient data, 15 subjects × 20 samples = 300 valid samples were retained for feature extraction and modeling. The balanced dataset improves generalization and reduces overfitting, providing a reliable foundation for the study.

II、 Signal Processing Framework

1. Overall Signal Processing Framework for Millimeter-Wave Radar-Based Blood Pressure Estimation

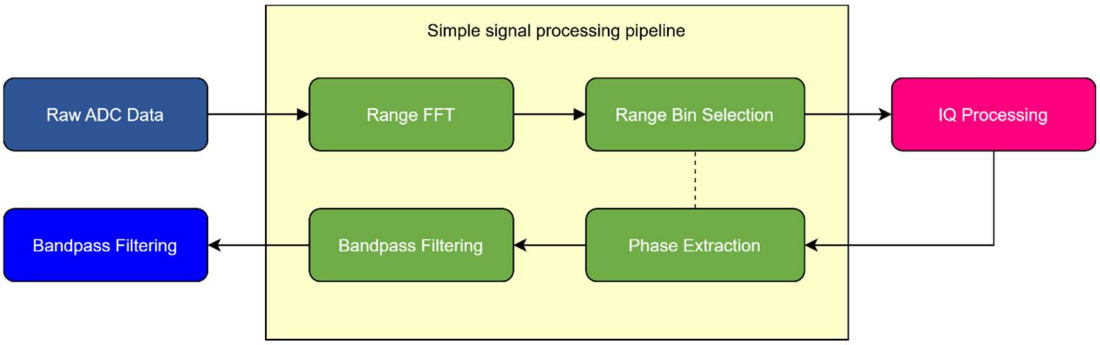

Fig 2 Framework for Millimeter-Wave Radar-Based Blood Pressure Estimation

2. Simple signal processing pipeline

The general signal processing pipeline is illustrated in Fig. 2.

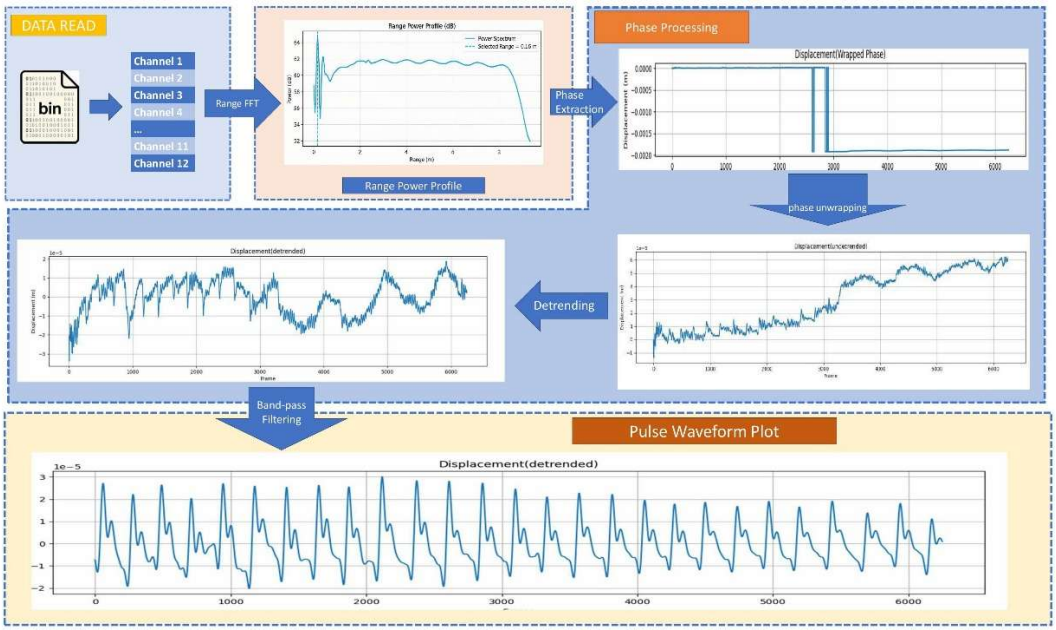

Fig 3 Simple signal processing

(1) Range FFT and Range Bin Selection

After performing range FFT on the complex echoes of the millimeter-wave radar, signal information corresponding to different range bins can be obtained.Using the 12-channel I/Q data, we compute the range spectrum via range FFT, then select the range bin with the highest signal power as the target range unit and extract its complex signal for subsequent processing.

(2) Phase Extraction

Phase extraction retrieves phase information from the target radar signal.Since the measured phase is periodic and discontinuous due to phase wrapping, phase unwrapping is applied to recover the true continuous phase variation, which reflects the micro-displacement of the target.

The signal is often contaminated by slow-varying trends caused by low-frequency phase drift and system imperfections such as local oscillator offset, I/Q imbalance, and multipath interference. These

trends do not correspond to real physiological micro-displacements.To eliminate this interference, detrending is performed by fitting and subtracting a low-order trend component from the phase signal.

(3) Bandpass Filitering

A Butterworth band-pass filter is adopted to suppress noise and extract effective pulse wave signals. Its frequency response is given by:

$$|H(j\omega)|^2 = \frac{1}{1+(\frac{\omega}{\omega_2})^{2n}} \cdot \frac{1}{1+(\frac{\omega_1}{\omega})^{2n}}$$

Where ω_1 and ω_2 denote the lower and upper cutoff angular frequencies, and n is the filter order. The passband is set to 0.5–5 Hz to retain valid pulse wave information.

The extracted waveform shows distinct pulse wave characteristics, yet DC offset remains uncorrected—this issue is addressed in the following section..

3. DC Offset Estimation and Removal

DC offset in millimeter-wave radar signals mainly originates from hardware non-idealities and static reflections, such as I/Q imbalance, LO leakage and ADC offset.It can be expressed as:The DC offset can generally be expressed as follows:

$$x(t) = I(t) + jQ(t) = (s_I(t) + I_0) + j\big(s_Q(t) + Q_0\big) \tag{8}$$

where I_0 and Q_0 denote the DC offset components, approximately constant over a short observation interval. Ideally, the IQ trajectory of a single dominant scatterer forms a circle centered at the origin, while hardware imperfections shift its center to (I_0, Q_0)

Circle fitting on the IQ trajectory allows estimation and compensation of the DC offset, thus improving the stability and accuracy of phase extraction.Among various circle center estimation methods, the RANSAC-based circle fitting algorithm is adopted in this work.

As a robust estimation approach, RANSAC iteratively fits candidate circles via random sampling and identifies the optimal circle model by distinguishing inliers from outliers, making it suitable for noisy IQ trajectory data.

Three non-collinear points are randomly sampled to estimate the circle parameters (x_c, y_c), The distance error between each data point and the fitted circle is then computed as d_i.

$$d_i = \left|\sqrt{(x_i - x_c)^2 + (y_i - y_c)^2} - R\right| \tag{9}$$

Points with $d_i < \epsilon$ (a predefined threshold) are classified as inliers.After multiple iterations, the model with the maximum number of inliers is chosen as the optimal fitting result.

Considering that the radar system employed in this study uses 3 transmitters and 4 receivers, forming a total of 12 channels, circle fitting is performed independently on the data from each channel.

(1) Direct circle fitting on ADC data

Since DC offset is typically superimposed directly on the digitized ADC output, it is reasonable to perform IQ constellation fitting on the raw complex data of the 12 channels.

The fitting results are shown in the figure below:

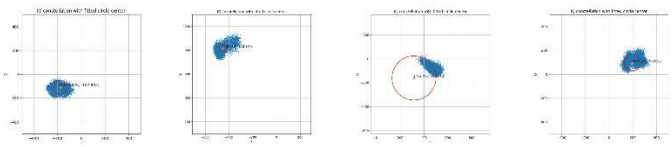

Fig 5 Circle fitting results.

The IQ constellations show no clear circular trajectories but only trend-like structures, resulting in poor phase rotation preservation, unstable circle fitting and large errors.

The IQ distributions of the 12 virtual channels deviate from the ideal circle due to strong noise and multipath interference, violating the single dominant scatterer assumption and degrading fitting performance.

Direct circle fitting on raw IQ data is thus unstable and sensitive to SNR and multipath effects. To solve this problem, a new DC offset removal method is proposed in this paper.

(2) IQ Circle Fitting Method Based on Range FFT-Derived Target Range Bin

As shown in Eq. (8), the received complex signal can be written as

$$x(t) = s_I(t) + js_Q(t) + (I_0 + jQ_0)$$

where $s_I(t)$and $s_Q(t)$are orthogonal components, and I_0, Q_0denote DC offsets. In the frequency domain, the DC term corresponds to an impulse at $f = 0$. However, due to finite-length DFT and windowing, spectral leakage spreads this energy to nearby frequencies. After range FFT and target bin selection, the signal can be approximated as

$$x(t) \approx s(t) + (I_0' + jQ_0')$$

Assuming a dominant scatterer, the ideal signal preserves its orthogonal structure and can be expressed as

$$s(t) = Ae^{j\phi(t)}$$

Thus, the received signal becomes

$$x(t) \approx Ae^{j\phi(t)} + (I_0' + jQ_0')$$

In the absence of DC offset, the I/Q components satisfy $I^2(t) + Q^2(t) = A^2$, forming a circle centered at the origin. With DC offset, this becomes

$$(I - I_0')^2 + (Q - Q_0')^2 = A^2$$

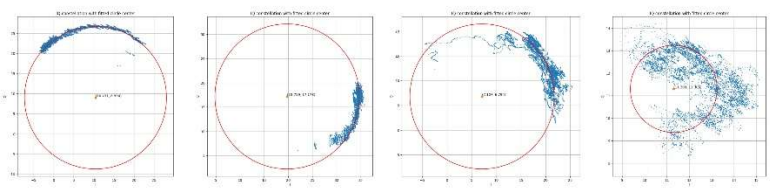

Fig 6 Circle fitting results.

The proposed method effectively restores the circular trajectory in the IQ plane, improving the stability and accuracy of circle fitting and enabling reliable DC offset estimation and removal for pulse wave extraction.

However, in some channels, high noise and multipath effects distort the IQ trajectory, leading to unstable fitting results (see Fig. 4). These channels typically lack clear pulse wave features and are unsuitable for analysis. Nevertheless, a subset of channels maintains sufficient signal quality, which is adequate to support subsequent pulse wave extraction.

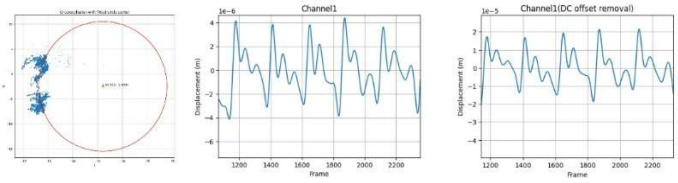

Fig. 7. Comparison before and after DC offset removal.

It can be observed that DC offset removal has a significant impact on pulse wave measurement, mainly reflected in changes in amplitude scale and waveform shape. The proposed method demonstrates effective performance.